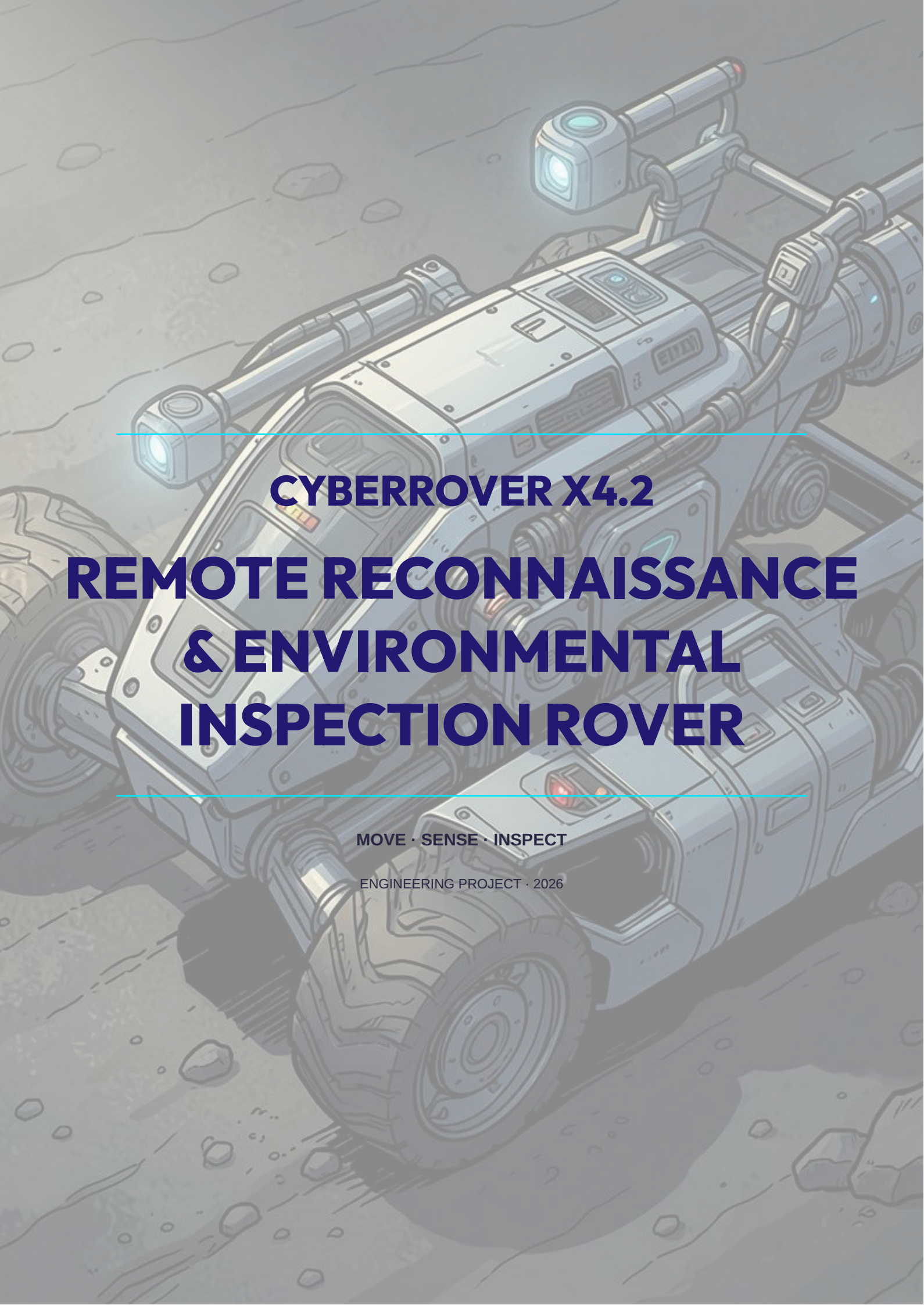




CYBERROVER X4.2

REMOTE RECONNAISSANCE & ENVIRONMENTAL INSPECTION ROVER

MOVE · SENSE · INSPECT

ENGINEERING PROJECT · 2026

WHEN THE ENVIRONMENT IS UNCERTAIN, DON'T SEND THE HUMAN FIRST.

Many environments present hazards, confinement, or inaccessibility that make direct human inspection impractical or dangerous. CyberRover X4.2 is a **prototype remote inspection platform** designed to extend the operator's senses into spaces where human presence carries unacceptable risk — gas-leak zones, confined spaces, fire-affected locations, earthquake-damaged structures, and industrial inspection areas.

OPERATIONAL FLOW

A four-stage proxy loop that keeps the human safe while delivering critical intelligence.



① HAZARDOUS AREA

Uncertain, inaccessible, or dangerous environment identified



② CYBERROVER X4.2

Prototype rover deployed into target zone



③ MOVE + SENSE + OBSERVE

Data collected, environment assessed, video streamed



④ REMOTE OPERATOR

Safe, informed decision-making from a distance

ONE PLATFORM. MULTIPLE SENSES.

The CyberRover X4.2 integrates four independent modular subsystems — each developed, tested, and upgradeable on its own terms. All control and interpretation remains with the remote human operator.

MOVE

4WD robotic mobility via differential skid-steering. Four geared DC motors with dual BTS7960 drivers for varied terrain.

SENSE

Gas detection, temperature, humidity, and barometric pressure/altitude mapping.

OBSERVE

Live visual feed via operator-mounted mobile camera. Remote visual inspection from the rover's on-site perspective.

CONNECT

A dedicated ESP32-CAM Wi-Fi Telemetry Hub broadcasts sensor data to the dashboard, keeping driving control offline and lag-free.

INTEGRATED CONTROL, SENSING & COMMUNICATION

① CONTROL PATH

Remote ESP32 → ESP-NOW → ESP32-S3
N8R16 → Arduino Uno → BTS7960 ×2 → 4×
DC Motors

② ENVIRONMENTAL PATH

MQ-4 / MQ-7 / MQ-135 / DHT22 → Arduino
Nano → UART Serial → ESP32-CAM
Telemetry Hub → Telemetry Dashboard
Arduino Nano → 16×2 LCD (Local Display)

③ ULTRASONIC PATH

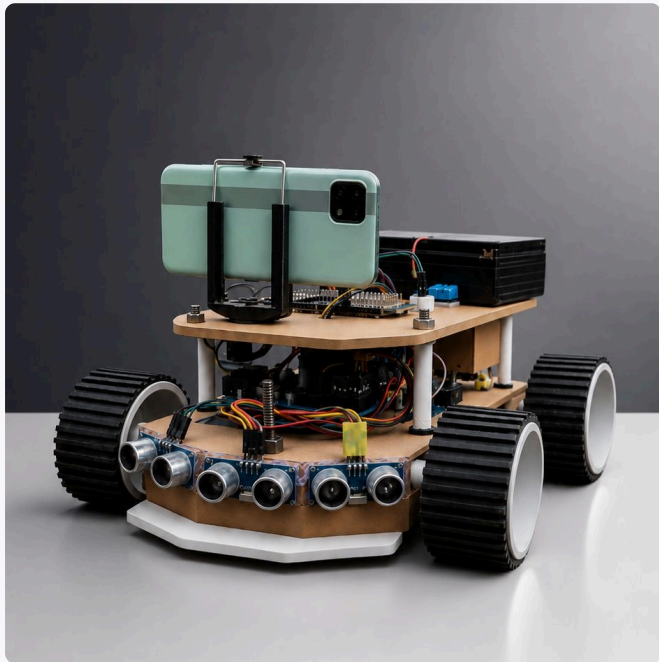
HC-SR04 ×3 (Left / Centre / Right) →
Arduino Uno → Serial / I²C → ESP32-S3
N8R16 → Telemetry Dashboard

④ TELEMETRY & VISUAL

ESP32-CAM Telemetry Hub → Wi-Fi (HTTP /
WebSocket) → Browser Dashboard
Phone Camera (Mounted) → Live Stream →
Operator Screen

BUILT TO MOVE THROUGH UNCERTAIN TERRAIN.

The physical design prioritizes stability, distributed electronics, practical sensor placement, and sufficient ground clearance for prototype-stage operation in varied environments.



KEY COMPONENTS

4WD Chassis

Structural foundation for all mounted subsystems.

Geared DC Motors

Four independent drive motors with torque multiplication.

BTS7960 Drivers

Dual modules for independent left/right motor control.

Electronics Platform

Central mount for ESP32-S3, Arduino Nano, and drivers.

Sensor Mounting

Ultrasonics at forward-left, center, and right; gas and DHT11 sensors distributed for ambient coverage.

Camera / Phone Mount

Forward-facing attachment for live visual inspection.

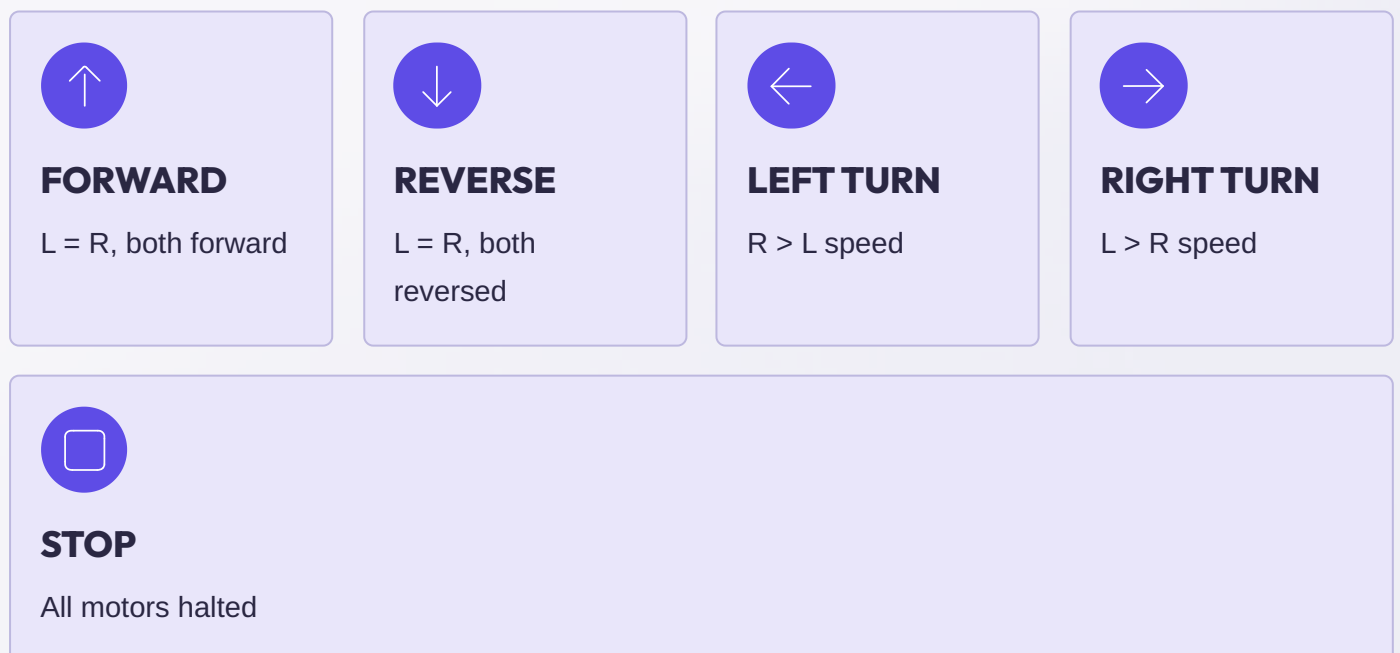
4WD DIFFERENTIAL MOBILITY

Four geared DC motors provide independent traction through all four wheels. The ESP32-S3 relays control commands to an **Arduino Uno**, which generates PWM signals for two BTS7960 H-bridge modules — enabling precise differential (skid) steering.

MOTOR CONTROL CHAIN



SKID-STEERING STATES



Skid-steering: Equal PWM → straight. Differential PWM between left and right groups → turning. No mechanical steering linkage required.

THREE DIRECTIONS. ONE DECISION.

Three ultrasonic sensors — positioned **left, center, and right** — scan the rover's forward hemisphere **sequentially, one at a time**. Distance data feeds the operator dashboard for informed movement decisions in confined or visually obscured spaces. This is sensor-assisted navigation — the operator retains full control at all times.

SENSOR POSITIONS

◀ LEFT

Forward-left detection

▲ CENTER

Primary forward detection

▶ RIGHT

Forward-right detection

i Sensors trigger and read **individually in sequence** — not simultaneously.

SEQUENTIAL DETECTION CYCLE



TRIGGER LEFT → READ LEFT

Pulse emitted; return time measured for left distance



TRIGGER CENTER → READ CENTER

Pulse emitted; return time measured for forward distance



TRIGGER RIGHT → READ RIGHT

Pulse emitted; return time measured for right distance



COMPARE DISTANCES

All three values checked against obstacle threshold



OPERATOR DECISION → DRIVE

Operator selects direction; motor command executed. Cycle restarts immediately.

ENVIRONMENTAL INSPECTION

The CyberRover X4.2 carries a suite of environmental sensors into locations where direct human presence may be unsafe. Raw sensor data is collected by the Arduino Nano, transmitted via UART to the ESP32-CAM, and streamed to the operator dashboard for preliminary environmental assessment.

SENSOR MODULES

MQ-4 — Combustible Gas

Methane & combustible gas indication. Raw ADC values. Pre-entry hazard assessment for gas-leak zones.

MQ-7 — Carbon Monoxide

CO presence indication. Raw ADC values. Fire-affected areas, enclosed spaces, post-combustion zones.

MQ-135 — Air Quality

Broad-spectrum volatile compound indication. Raw ADC values. General atmospheric hazard screening.

DHT11 & BMP280 — Temp, Humidity, Pressure & Altitude

Ambient temperature (°C), relative humidity (%), barometric pressure (hPa), and altitude (m). Standard engineering units. Thermal, moisture, and atmospheric context.

DATA PATH



SENSORS

MQ-4 · MQ-7 · MQ-135 · DHT11 · BMP280



ARDUINO NANO

Reads & digitizes sensor outputs



UART

Serial data transmission link



ESP32-CAM TELEMETRY HUB

Transmits data to operator dashboard



Gas sensors output raw ADC values only. No certified ppm concentrations are calculated. Readings support **preliminary** environmental assessment — not definitive hazard classification.

FROM SENSOR TO OPERATOR.

All sensor and control data is transmitted wirelessly from the CyberRover X4 to the operator laptop in real-time. Raw readings are routed through two microcontrollers, serialized over UART, and broadcast over Wi-Fi to the operator dashboard.

TELEMETRY PIPELINE

SENSORS MQ-4 · MQ-7 · MQ-135 · DHT11 · BMP280 · Ultrasonic ×3	ARDUINO NANO / UNO Reads & digitizes all sensor outputs	UART Serial data link to ESP32-CAM	ESP32-CAM TELEMETRY HUB Aggregates & transmits over Wi-Fi	DASHBOARD Live operator display
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OPERATOR DASHBOARD — DISPLAY PANELS

ENVIRONMENT

MQ-4, MQ-7, MQ-135 raw ADC gas readings

DISTANCE

Left, Center, Right ultrasonic proximity channels

TEMPERATURE

DHT11 ambient temperature in °C

ROBOT STATUS

Live connection state and control indicator

HUMIDITY

DHT22 relative humidity in % RH

VISUAL

Phone-camera live observation stream



Gas readings are raw ADC values — no ppm concentrations calculated. Operators interpret all data alongside the live visual feed to assess conditions.